

Iron Triangle — Evidence Appendix

Items S1–S3 and E1–E5

All times are elapsed model hours from $t = 0$ (2026-09-22 00:00:00 TDB). All traces are conditional no-loss unless labeled otherwise. Probabilities are reported with their dependencies. The opening balance sheet is Table 1. E4 and E5 state the tier claimed. The initiator for each transaction is elected by minimising $T(A) = d(S_1, A) + 2 \min_{B \neq A} d(A, B) + d(A, S_2)$, with ties broken by lowest node ID (Mercury 1, Venus 2, Earth 3). In the five baseline scenarios below, Earth is one of the two parties in each trade, so Earth is elected in every trace; the rotating rule matters most for inner-system pairs.

1 Opening Balance Sheet

Table 1: Opening balance sheet. Eight accounts at eight settlements. Totals: \$500,000 and 5,000 shares.

Account	Settlement	NeoDollars	Shares	Notes
Terra Capital LLC	Earth	150,000	600	Anchor client
Mercury Solar Energy	Mercury	90,000	900	Anchor client
Venus Ventures	Venus	80,000	800	Anchor client
Mars Manufacturing	Mars	70,000	700	Non-anchor client
Ceres Aggregate	Ceres	40,000	400	Non-anchor client
Jupiter Gateway Hldgs	Jupiter	20,000	200	Outer client
Saturn Ring Logistics	Saturn	25,000	500	Outer client
Neptune Deep Resources	Neptune	25,000	900	Farthest client
Total		500,000	5,000	

Conventions. “Anchor” is one of Earth, Mercury, Venus. “Quorum” is any two of the three anchors. “Spendable” means the winner can use the amount as a balance at the winner’s own settlement after local apply. Every packet counts unless a row is marked quota-exempt. Where two copies share a closure or forced-loss window we say so and do not multiply.

2 S1. Scenario Traces

2.1 Scenario A: Value move Earth to Neptune

A client at Earth (Terra Capital) buys 50 Ares Habitat shares from Esha at Neptune (Neptune Deep Resources) at \$100/share. Notional \$5,000. Earth is one party, so Earth is the elected initiator; fee \$0.50 to the quorum.

Table 2: Scenario A conditional no-loss trace. Elected initiator: Earth.

h	Actor / knowledge	Action	Packet	State after
0.0000	Buyer and seller know only home	Local orders; Earth locks \$5k; Neptune locks 50 shares; Earth balances.	Local.	Both legs locked.
0.0006	Earth anchor has buyer's lock.	PREPARE to Mercury and Venus.	Backbone E-A-Mer, E-A-Ven.	Pending.
0.0500	Mercury anchor receives PREPARE.	COMMIT.	Backbone Mer-A-E.	Replica updated.
0.0800	Venus anchor receives PREPARE.	COMMIT.	Backbone Ven-A-E.	Replica updated.
0.1300	Earth anchor has both votes.	COMMIT to Mercury and Venus.	Backbone E-A-Mer, E-A-Ven.	Trade final at Earth.
0.1800	Mercury anchor records final.	Local.	—	Replica final.
0.2100	Venus anchor records final.	Local.	—	Replica final.
250.40	Neptune gateway receives COMMIT.	Release to seller.	Local access.	Seller spendable at Neptune.
250.42	Earth gateway receives COMMIT.	Release to buyer.	Local access.	Buyer spendable at Earth.

Completion time. Spendable moment 250.42 h. Discharge 0.130 h. Backed-claim 0.180 h (Mercury) and 0.210 h (Venus). **Communications.** Backbone 6 launches; direct 2 local access; total 6. Quota: 6/600. **Peak encumbered.** \$5,000. **Asset-hours.** $5,000 \times 250.42 = 1,252,100$ NeoDollar-hours. **Exposure.** Each side \$5,000. **Capital utilisation** $5,000/500,000 = 1.0\%$.

2.2 Scenario B: Long-lived obligation (500-hour Mars bond)

Mars Manufacturing (Mars) issues a 500-hour bond, \$150,000 principal, 5% coupon at maturity. Terra Capital (Earth) buys at par. Terra's principal is escrowed for the term; Mars Manufacturing pledges 700 Ares shares as coupon and default backing. Elected initiator: Earth. The bond is final at $h = 0.1303$ and matures at $h = 500$.

Table 3: Scenario B completion.

Quantity	Value
Elected initiator	Earth
Discharge	$h = 500.00$
Spendable	$h = 500.30$
Backbone launches	6
Direct/local	2 local access
Peak encumbered	\$150,000 cash; 700 shares
Asset-hours (cash)	$150,000 \times 500.30 = 75,045,000$
Asset-hours (shares)	$700 \times 500.30 = 350,210$
Capital utilisation	$150,000/500,000 = 30.0\%$

2.3 Scenario C: Price-rising run

Cash-settled future on Ares, 10 contracts, multiplier 100, entry 100. Notional \$100,000. Margin is 30% of notional = \$30,000 each side. Final observation at maturity is $P = 135$, which exceeds the $\pm 30\%$ cap, so settlement is capped at effective price 130. The long side receives $10 \times 100 \times 30 = \$30,000$ from the short's posted margin. Elected initiator: Earth.

Table 4: Scenario C conditional no-loss trace.

h	Actor / knowledge	Action	Packet	State after
0.0000	Buyer (Earth) submits.	Local orders; Amina locks \$30k at Earth; Bayo locks \$30k at Mars; Earth elected.	Local.	Both margins locked; position pending.
0.1300	Earth anchor has both votes.	COMMIT.	Backbone.	Position open.
240.00	PBS releases final $P=135$.	Direct $V \rightarrow E$.	Direct.	Packet in flight.
240.30	Earth anchor marks to 135; cap applies; effective price 130.	Local.	—	Margin covered; no variation required.
480.00	Quorum matures.	Pay long \$30,000 from short's margin.	Backbone.	Discharge.
480.30	Buyer spendable at Earth; seller's remaining margin released at Mars.	Local.	—	Closed.

Completion time. Spendable 480.30 h. **Communications.** Backbone 6; direct 2; total 8. **Peak encumbered** \$60,000 cash (\$30,000 each side). **Asset-hours** $30,000 \times 480.30 \times 2 = 28,818,000$ cash-h. **Exposure** \$30,000 per side. **Capital utilisation** $60,000/500,000 = 12.0\%$.

2.4 Scenario D: Price-falling run

Same contract, entry 100, final price 65. The $\pm 30\%$ cap applies, so the long pays \$30,000 to the short. Because the cap equals the initial margin, the posted margin already covers the maximum possible loss and *no margin call is required*. Elected initiator: Earth.

Table 5: Scenario D conditional no-loss trace.

h	Actor / knowledge	Action	Packet	State after
0.0000	Seller (Earth) submits.	Local orders; Amina locks \$30k at Earth; Bayo locks \$30k at Mars; Earth elected.	Local.	Both margins locked; position pending.
0.1300	Earth anchor has both votes.	COMMIT.	Backbone.	Position open.
240.00	PBS releases final $P=65$.	Direct $V \rightarrow E$.	Direct.	Packet in flight.
240.30	Earth anchor marks to 65; cap applies; effective price 70.	Local.	—	Margin already covers cap; no call.
480.00	Quorum matures.	Pay short \$30,000 from long's margin.	Backbone.	Discharge.
480.30	Seller spendable at Mars; long's remaining margin released at Earth.	Local.	—	Closed.

Completion time. Spendable 480.30 h. **Communications.** Backbone 6; direct 2; total 8. **Peak encumbered** \$60,000 cash (\$30,000 each side). **Asset-hours** $30,000 \times 480.30 \times 2 = 28,818,000$ cash-h. **Exposure** \$30,000 per side. **Capital utilisation** $60,000/500,000 = 12.0\%$.

2.5 Scenario E: Equity trade (second product kind)

Terra Capital (Earth) sells 300 Ares Habitat shares to Venus Ventures (Venus) at \$100/share. Notional \$30,000. Fee is 1 bp = \$3, paid by the buyer to the quorum. Elected initiator: for an Earth–Venus pair at $h = 0$, $T(\text{Earth}) = 2.028$ h, $T(\text{Venus}) = 2.308$ h, $T(\text{Mercury}) = 3.215$ h; Earth is elected. Using the $h = 0$ Venus–Earth route delay of 40.568 min one way, Venus receives PREPARE at about $h = 0.676$, its COMMIT reaches Earth at about $h = 1.352$, and Earth's COMMIT reaches Venus at about $h = 2.028$. The spendable moment is the later local apply, about $h = 2.03$.

Table 6: Scenario E conditional no-loss trace. Elected initiator: Earth.

h	Actor / knowledge	Action	Packet	State after
0.0000	Buyer and seller know only home balances.	Local orders; Earth locks 300 shares; Venus locks \$30k; Earth elected.	Local.	Both legs locked.
0.0006	Earth anchor has seller's lock.	PREPARE to Mercury and Venus.	Backbone E-A-Mer, E-A-Ven.	Pending.
0.0500	Mercury anchor receives PREPARE.	COMMIT.	Backbone Mer-A-E.	Replica updated.
0.6760	Venus anchor receives PREPARE.	COMMIT.	Backbone Ven-A-E.	Replica updated; quorum has two votes once Earth receives it.
1.3520	Earth anchor has Earth + Mercury + Venus votes.	COMMIT to Mercury and Venus.	Backbone E-A-Mer, E-A-Ven.	Trade final at Earth; seller spendable at Earth.
1.7000	Mercury anchor records final.	Local.	—	Replica final.
2.0280	Venus anchor receives signed COMMIT and applies once.	Local apply.	Backbone E-A-Ven.	Buyer spendable at Venus; trade complete.

Completion time. Spendable moment 2.028 h. Discharge 1.352 h. Backed-claim 2.028 h (Venus). **Communications.** Backbone 6 launches; direct 2 local access; total 6. Quota: 6/600. **Peak encumbered.** \$30,000 cash and 300 shares. **Asset-hours.** $30,000 \times 2.028 = 60,840$ NeoDollar-hours; $300 \times 2.028 = 608.4$ share-hours. **Exposure.** Each side \$30,000. **Capital utilisation.** $30,000/500,000 = 6.0\%$.

End balances. Terra Capital: cash \$180,000; shares 300. Venus Ventures: cash \$49,997; shares 1,100. Quorum (via fee): \$3. All other accounts unchanged. Total cash \$500,000; total shares 5,000.

3 S2. Stress and Recovery

3.1 Incident choice

Incident. Earth gateway isolation for 72 h, starting at $h = 230$ and ending at $h = 302$. This is a damaging allowed incident for the Triad because Earth is both a busy initiator and the destination of a large share of demand. During isolation, Earth-side cells cannot originate new cross-cell transactions and Earth cannot serve as an initiator, but Mercury and Venus continue to serve cross-cell work between non-Earth parties at a two-of-three quorum. Existing Earth-side locks remain durable.

3.2 Stress timeline (Refer to Table 7)

Duration of lost service. 72.00 h for Earth-side origination. 0 h for non-Earth work that can form the Mercury–Venus quorum; a Mars–Neptune transfer, for example, is initiated by Venus with Mercury voting and settles without Earth. **Recovery** at 303.10 h for full three-anchor service, 1.10 h after isolation ends. **State of every obligation.** All final before 230.00 remain final on all three anchors. All submitted during [230, 302) that touch Earth are held; those that do not touch Earth are handled by the Mercury–Venus quorum. No obligation is cancelled or reset.

3.3 Recovery after connectivity returns

Earth syncs from Mercury and Venus journals, adopts their committed records, and resumes as a fully available initiator. Pending work is processed in time order. Each pending obligation needs one prepare round (two backbone launches) and one commit round (two backbone launches). Full backlog drains within about 2.5 h.

Table 7: S2 stress run: Earth gateway isolated $h \in [230, 302]$.

Time	Event / knowledge	Financial state
99.99	Normal.	All final records on three anchors.
230.00	Earth isolation begins.	Existing records preserved; local and direct only for Earth-side work.
230.10	Mercury + Venus confirm mutual quorum.	Non-Earth work continues at two-of-three quorum.
240.565	PBS final P=\$130 at Earth; direct advisory to Mars lost. Mercury + Venus sign the final outcome at quorum.	Amina’s local Earth margin handled at Earth; Bayo’s \$75k margin stays locked at Mars until quorum commit arrives.
265.7, 290.8	$R_e = 2T_0 + 24$ h; two retries fall inside isolation.	Same quorum; no duplicate payoff, no release of Mars hold.
302.00	Isolation ends; routes via Relay A clear.	Fresh workflows resume; Earth syncs from Mercury/Venus.
303.10	First new two-phase spot completion under full three-anchor quorum.	Full service restored.
316.50	Old direct-email to Earth’s primary register is rejected as an outdated view.	Original liability already discharged by the quorum; no double effect.

3.4 Maintenance comparison

Table 8: S2 maintenance comparison.

Run	First new commit (h)	Lost service (h)	Recovery delay (h)
With scheduled maintenance	303.10	72.00	1.10
Without scheduled maintenance	303.05	72.00	1.05
Difference	0.05	0.00	0.05

3.5 Sensitivity: no rotating initiator

If only Earth could initiate, Earth isolation would pause all new cross-cell commits until 302.00. The rotating rule reduces lost financial service for non-Earth cross-cell work from 72 h to 0 h and reduces full-service recovery from 72 h to about 1.1 h.

4 S3. Access at Every Settlement

4.1 Access tables at $h = 0$ and $h = 300$

Table 9: S3 access tables, exact Kepler geometry. Left: $h = 0$. Right: $h = 300$.

Settlement	Best route	Delay (min)	Avail.	Settlement	Best route	Delay (min)	Avail.
Mercury	Local (anchor)	0.017	1.00	Mercury	Local (anchor)	0.017	1.00
Venus	Local (anchor)	0.017	1.00	Venus	Local (anchor)	0.017	1.00
Earth	Local (anchor)	0.017	1.00	Earth	Local (anchor)	0.017	1.00
Mars	Mars–A–Earth	33.907	1.00	Mars	Mars–A–Earth	33.439	1.00
Ceres	Ceres–A–Earth	33.370	1.00	Ceres	Ceres–A–Earth	32.244	1.00
Jupiter	Jupiter–B–Earth	50.923	1.00	Jupiter	Jupiter–B–Earth	50.229	1.00
Saturn	Saturn–A–Earth	79.419	1.00	Saturn	Saturn–A–Earth	78.862	1.00
Uranus	Uranus–A–Earth	158.255	1.00	Uranus	Uranus–A–Earth	156.719	1.00
Neptune	Neptune–A–Earth	250.480	1.00	Neptune	Neptune–A–Earth	250.052	1.00

All selected routes sample at 100% at both snapshots. Every settlement can reach the elected initiator within its own row’s delay; the access table doubles as the input to the path-election rule. For any pair (S_1, S_2) the originating cell evaluates $T(A)$ for $A \in \{\text{Earth}, \text{Mercury}, \text{Venus}\}$ and sends to the minimum. In all five baseline scenarios Earth is a party, so Earth is elected; the table is shown for local-anchor access to keep the traces directly comparable.

4.2 No-loss value-move timeline for three settlements

Table 10: S3 no-loss value-move timeline. Same Mars seller, same 100-share Ares trade, re-run with Amina’s home changed.

Amina’s home	Amina → anchor (min)	Mars → anchor (min)	Completion (h)	Availability
Earth (best)	0.017	33.907	1.130	1.00
Mercury (median)	0.017	33.907	1.130	1.00
Neptune (worst)	250.480	33.907	8.349	1.00

The completion column is $2\times$ the slower participant-to-anchor one-way route, rounded, for prepare and commit; it includes no random loss. In each re-run Amina starts with \$150,000 and 600 shares at the named site; Bayo’s opening account remains Mars. For each, \$10,000 / 100 shares are locked, fee is \$1, and end totals equal the \$1 spot run.

4.3 Worked initiator election

Table 11: Initiator election for representative pairs at $h = 0$.

Pair (S_1, S_2)	$T(\text{Earth})$ (h)	$T(\text{Mercury})$ (h)	$T(\text{Venus})$ (h)	Elected
Earth–Venus	2.028	3.215	2.308	Earth
Mercury–Venus	1.380	0.086	0.086	Mercury (tie, lowest ID)
Earth–Mars	1.130	2.609	2.426	Earth
Mars–Neptune	9.491	8.496	8.323	Venus
Ceres–Jupiter	1.850	2.706	2.528	Earth

The Mars–Neptune election illustrates the rotating rule: Venus wins by 0.17 h over Mercury and 1.17 h over Earth. The Mercury–Venus election illustrates the tie-break: the two anchors are identical by symmetry, so the lowest node ID wins. Under an Earth-only rule, the Mercury–Venus pair would pay 1.38 h instead of 0.086 h, a savings of about 1.3 h per inner-system transaction.

5 E1. Orbital Calculations

5.1 Propagation method

We use the supplied `orbital_elements.json` fixed Kepler elements, exact $M(t) = M_0 + n t$ in days modulo 360° , Newton solution of $E - e \sin E = M$, orbital-plane position

$$x' = a(\cos E - e), \quad y' = a\sqrt{1 - e^2} \sin E, \quad z' = 0,$$

and rotation into the Sun-centered J2000 ecliptic frame

$$\mathbf{r} = R_z(\Omega) R_x(i) R_z(\omega) \mathbf{r}'.$$

Relays A and B use the circular orbits in `network_model.json`: $a = 2.8284271247461903$ AU, $i = 0$, phases 45° and 135° at epoch, period 1737.4644090487413 d.

5.2 Epoch check

Table 12: E1 epoch positions from the code, matching the brief to $< 10^{-5}$ AU.

Body	x (AU)	y (AU)	z (AU)
Mercury	-0.213426	-0.410281	-0.013955
Venus	0.679067	-0.257950	-0.042725
Earth	1.003581	-0.023693	-0.000003
Mars	0.247476	1.526869	0.025929
Ceres	0.375786	2.653941	0.014791
Jupiter	-3.438179	4.038309	0.060149
Saturn	9.271221	1.717485	-0.398962
Uranus	8.962475	17.253919	-0.052129
Neptune	29.839098	1.351785	-0.715470
Relay A	2.000000	2.000000	0.000000
Relay B	-2.000000	2.000000	0.000000

5.3 Light-time solution

We solve

$$\tau = \frac{|\mathbf{r}_{\text{recv}}(t_e + \tau) - \mathbf{r}_{\text{send}}(t_e)|}{c}, \quad c = 173.1446 \text{ AU/day},$$

by fixed-point iteration from $\tau = 0.01 \text{ d}$ to tolerance 10^{-7} d . Times are converted to minutes at 8.317 min/AU for reporting.

5.4 Worked receiver-motion example

Earth to Mars at the epoch. At emission, simultaneous endpoints are 1.725284793 AU apart, giving $8.317 \times 1.725284793 = 14.34919363 \text{ min}$ if Mars were frozen. Iterating Mars to its arrival position gives 1.725373560 AU and 14.34993186 min of light flight: receiver motion adds 0.0443 s . The return packet starts at the forward arrival and is solved anew: 1.725067713 AU and 14.34738806 min . We do not reuse the forward delay.

5.5 Solar exclusion

The exclusion test uses the minimum distance of the emitted-to-arrival straight segment from the Sun, not a same-time line. For an Earth–Relay A ray entering occultation, the refined first blocked emission is $h = 6667.940104 \text{ h}$ (bracket $[6667.94010, 6667.94011]$, width 0.036 s). A packet emitted just before the boundary has closest miss 0.1000001 AU and remains in flight; a new launch during the blocked interval is unavailable.

6 E2. Probability and Traffic

6.1 Direct-service success probabilities

For n independent copies each failing with probability p , all fail with p^n ; success is $1 - p^n$. Where copies share a closure we do not multiply.

- Venus observation to Earth: $d \approx 1.3 \text{ AU}$, $p = 1 - e^{-0.08 \times 1.3} = 0.0989$. Single-copy success 0.9011 ; two-copy 0.9902 .
- Venus observation to Mercury: $d \approx 0.8 \text{ AU}$, $p = 0.0620$. Single-copy 0.9380 ; two-copy 0.9962 .
- Neptune to Earth direct: $d \approx 30 \text{ AU}$, $p = 0.9093$. Single-copy success 0.0907 ; the client uses the backbone.
- Worst direct bound: $d_{\text{max}} = 1.000 + 30.069 = 31.069 \text{ AU}$, $p_{\text{max}} = 0.9167$, single-copy success 0.0833 .

6.2 Backbone hop probabilities

For a hop with up to four launches, each failing with probability p , all four fail with p^4 . Using median backbone loss from the 2000-year scan:

- Mercury–Relay A: median 0.0406%; all-four-fail 2.7×10^{-14} .
- Relay A–Neptune: median 0.4579%; all-four-fail 4.4×10^{-10} .
- Full Mercury–Neptune route: route abandonment $\leq 4.4 \times 10^{-10}$.
- Earth–Mercury route: median 0.0409% per hop; route abandonment below 10^{-16} .

6.3 Packet counts

Table 13: E2 packet counts by scenario.

Scenario	Backbone	Direct	Total	Quota
A (value move)	6	2	8	6/600 backbone; 2/12 direct
B (bond)	6	2	8	6/600; 2/12
C (rising)	6	2	8	6/600; 2/12
D (falling)	6	2	8	6/600; 2/12
E (equity)	6	2	8	6/600; 2/12
S2 (stress)	8	4	12	8/600; 4/12

Each directed link carries at most 1 packet/s. Our scenarios use at most 12 packets over hundreds of hours. The 10,000-packet queue cap is not approached.

7 E3. Balances and Capital

7.1 Balances and obligations over time (Refer to Table 14)

Table 14: E3 balance sheet at hour 0.

Asset	Owner	Location	Amount	Encumbrance
NeoDollars	Terra Capital	Earth	150,000	0
NeoDollars	Mercury Solar	Mercury	90,000	0
NeoDollars	Venus Ventures	Venus	80,000	0
NeoDollars	Mars Manufacturing	Mars	70,000	0
NeoDollars	Ceres Aggregate	Ceres	40,000	0
NeoDollars	Jupiter Gateway	Jupiter	20,000	0
NeoDollars	Saturn Ring	Saturn	25,000	0
NeoDollars	Neptune Deep	Neptune	25,000	0
Shares	Terra Capital	Earth	600	0
Shares	Mercury Solar	Mercury	900	0
Shares	Venus Ventures	Venus	800	0
Shares	Mars Manufacturing	Mars	700	0
Shares	Ceres Aggregate	Ceres	400	0
Shares	Jupiter Gateway	Jupiter	200	0
Shares	Saturn Ring	Saturn	500	0
Shares	Neptune Deep	Neptune	900	0

7.2 Time series for Scenario A (Refer to Table 15)

Table 15: E3 time series for Scenario A.

Hour	Encumbered \$	Free \$	Note
0.000	0	500,000	Before any action
0.0006	5,000	495,000	Buyer's cash locked at Earth
0.0500	10,000	490,000	Seller's shares locked at Neptune (50 shares, \$5k equivalent)
0.1300	10,000	490,000	Trade final at Earth
250.40	5,000	495,000	Seller spendable; seller's encumbrance released
250.42	0	500,000	Buyer spendable; all released

7.3 Time series for Scenario B

Mars Manufacturing (Mars) issues a \$150,000 500-hour bond at 5% coupon to Terra Capital (Earth). Terra's principal is escrowed for the term; Mars Manufacturing pledges 700 Ares shares as coupon and default backing. Peak encumbrance is \$150,000 cash and 700 shares. At maturity, Terra receives principal plus \$7,500 coupon; Mars's 700 shares are released.

Table 16: E3 time series for Scenario B.

Hour	Encumbered \$	Encumbered shares	Free \$	Note
0.0000	0	0	500,000	Before any action.
0.0006	150,000	700	350,000	Terra locks \$150k as bond principal in escrow; Mars Manufacturing locks 700 shares as coupon/default backing.
0.1300	150,000	700	350,000	Bond final at the quorum.
500.00	150,000	700	350,000	Maturity: escrow releases \$150,000 to Terra; Mars Manufacturing pays \$7,500 coupon from its cash.
500.30	0	0	500,000	Terra spendable at Earth; all encumbrances released.

End balances. Terra Capital: cash \$157,500; shares 600. Mars Manufacturing: cash \$62,500; shares 700. All other accounts unchanged. Total cash \$500,000; total shares 5,000.

Peak encumbered. \$150,000 cash and 700 shares. **Asset-hours.** $150,000 \times 500.30 = 75,045,000$ cash-h; $700 \times 500.30 = 350,210$ share-h. **Exposure.** Terra \$150,000 principal at risk; Mars Manufacturing \$7,500 coupon and 700 shares at risk. **Capital utilisation.** $150,000/500,000 = 30.0\%$.

7.4 Time series for Scenario C

Cash-settled future on Ares, 10 contracts, multiplier 100, entry 100. Notional \$100,000. Margin is 30% notional = \$30,000 each side. Amina (Terra Capital, Earth) is long; Bayo (Mars Manufacturing, Mars) is short. Final observation $P = 135$, capped at +30% effective price 130; long receives \$30,000 from short's margin.

Table 17: E3 time series for Scenario C.

Hour	Encumbered \$	Free \$	Note
0.0000	0	500,000	Before any action.
0.0006	60,000	440,000	Amina locks \$30k at Earth; Bayo locks \$30k at Mars.
0.1300	60,000	440,000	Position open at the quorum.
240.00	60,000	440,000	PBS releases final $P = 135$; cap applies; effective price 130. Margin covered; no variation required.
480.00	60,000	440,000	Maturity: quorum moves \$30,000 from Bayo's margin to Amina.
480.30	0	500,000	Amina spendable at Earth; Bayo's remaining margin released at Mars. All encumbrances released.

End balances. Terra/Amina: cash \$180,000; shares 600. Mars/Bayo: cash \$40,000; shares 700. All other accounts unchanged. Total cash \$500,000; total shares 5,000.

Peak encumbered. \$60,000 cash. **Asset-hours.** $30,000 \times 480.30 \times 2 = 28,818,000$ cash-h. **Exposure.** \$30,000 per side (cap = margin; max loss = posted margin). **Capital utilisation.** $60,000/500,000 = 12.0\%$.

7.5 Time series for Scenario D

Same contract and same margin. Final observation $P = 65$, capped at -30% effective price 70; long pays \$30,000 from its margin to the short. Because the cap equals the margin, the posted margin covers the maximum possible loss and no margin call is required.

Table 18: E3 time series for Scenario D.

Hour	Encumbered \$	Free \$	Note
0.0000	0	500,000	Before any action.
0.0006	60,000	440,000	Amina locks \$30k at Earth; Bayo locks \$30k at Mars.
0.1300	60,000	440,000	Position open at the quorum.
240.00	60,000	440,000	PBS releases final $P = 65$; cap applies; effective price 70. Margin already covers cap; no call.
480.00	60,000	440,000	Maturity: quorum moves \$30,000 from Amina's margin to Bayo.
480.30	0	500,000	Bayo spendable at Mars; Amina's remaining margin released at Earth. All encumbrances released.

End balances. Terra/Amina: cash \$120,000; shares 600. Mars/Bayo: cash \$100,000; shares 700. All other accounts unchanged. Total cash \$500,000; total shares 5,000.

Peak encumbered. \$60,000 cash. **Asset-hours.** $30,000 \times 480.30 \times 2 = 28,818,000$ cash-h. **Exposure.** \$30,000 per side. **Capital utilisation.** $60,000/500,000 = 12.0\%$.

7.6 Time series for Scenario E

Terra Capital (Earth) sells 300 Ares Habitat shares to Venus Ventures (Venus) at \$100 per share. Notional \$30,000. Fee is 1 bp = \$3, paid by the buyer to the quorum. Earth and Venus lock their legs at $h = 0$; Earth is elected initiator. Completion is the later of Earth apply and Venus apply. Using the $h = 0$ Venus–Earth route delay of 40.568 min one way, the spendable moment is about $h = 2.03$.

Table 19: E3 time series for Scenario E.

Hour	Encumbered \$	Encumbered shares	Free \$	Note
0.0000	0	0	500,000	Before any action.
0.0006	30,000	300	470,000	Venus locks \$30k; Terra locks 300 shares.
0.6760	30,000	300	470,000	Venus receives PREPARE; replies COMMIT.
1.3520	30,000	300	470,000	Earth receives Venus COMMIT; quorum reached; Earth applies locally.
2.0280	0	0	500,000	Venus receives COMMIT and applies; buyer spendable.

End balances. Terra: cash \$180,000, shares 300. Venus Ventures: cash \$49,997, shares 1,100. Quorum: \$3. All other accounts unchanged. Total cash \$500,000; total shares 5,000.

Peak encumbered. \$30,000 cash and 300 shares. **Asset-hours.** $30,000 \times 2.028 = 60,840$ NeoDollar-hours; $300 \times 2.028 = 608.4$ share-hours. **Exposure.** Each side \$30,000. **Capital utilisation.** $30,000/500,000 = 6.0\%$.

7.7 Capital measures

Table 20: E3 capital measures by scenario.

Scenario	Peak enc. \$	Peak enc. shares	Asset-hours	Capital eff.
A	5,000	0	$5,000 \times 250.42 = 1,252,100$ cash-h	0.010
B	150,000	700	$150,000 \times 500.30 = 75,045,000$ cash-h; $700 \times 500.30 = 350,210$ share-h	0.300
C	60,000	0	$30,000 \times 480.30 \times 2 = 28,818,000$ cash-h	0.120
D	60,000	0	$30,000 \times 480.30 \times 2 = 28,818,000$ cash-h	0.120
E	30,000	300	$30,000 \times 2.028 = 60,840$ cash-h; $300 \times 2.028 = 608.4$ share-h	0.060

Scale. Scenario B: $150,000/500,000 = 30.0\%$ of opening NeoDollars encumbered at peak, above 25%. **Value settled.** A: \$5,000. B: \$150,000. C: \$100,000 notional. D: \$100,000 notional. E: \$30,000. **Binding variation.**

Raise Scenario C to 31 lots. Required margin $0.30 \times 31 \times 100 \times 100 = \$93,000$ each. Amina can post; Bayo has only \$70,000 free at Mars. The quorum rejects before open; no margin is debited, no position exists.

8 E4. Long-Horizon Scan

Tier claimed: **Tier 3**. The 200-year screen uses 1-hour ticks, $t = 0$ through 1,753,200 h inclusive: 1,753,201 samples. It evaluates all 19 fixed candidate links and the allowed simple routes to each anchor using exact retarded geometry. Every non-Earth settlement has a candidate route to at least one anchor at every sampled tick (100% sampled point reachability); anchors are local at 1 s.

8.1 Refined event boundary

The 1-hour screen brackets an entering Earth-to-Relay-A solar occultation at $h \in [6667, 6668]$. Re-solving the moving receiver and the closest point of the complete emitted-to-arrival ray gives first blocked emission at $h = 6667.940104$ (bracket $[6667.94010, 6667.94011]$, 0.036 s wide; required tolerance 0.1 s). The interval ends at about $h = 7161.404475$. It lasts about 493.464 h (20.56 d). A packet emitted just before the boundary remains in flight; a new launch during the blocked interval is unavailable.

8.2 Link availability and route-delay ranges

Table 21: E4 Tier 3. Coarse sampled route-delay ranges to nearest anchor and slow points.

Settlement	Sampled route-delay range (h)	Sampled slow point
Mercury	0.581–1.400	1.400 at $h = 1,007,527$
Venus	0.545–1.409	1.409 at $h = 284,315$
Mars	0.419–1.459	1.459 at $h = 1,539,743$
Ceres	0.285–1.530	1.530 at $h = 419,791$
Jupiter	0.551–1.837	1.837 at $h = 120,322$
Saturn	1.116–2.415	2.415 at $h = 206,700$
Uranus	2.411–3.791	3.791 at $h = 1,262,565$
Neptune	3.994–5.194	5.194 at $h = 650,213$

8.3 Blinded-window analysis

Over the 2000-year scan, Earth and Mercury align every 115.88 days, about 6,300 times. For each alignment we check whether Earth–Neptune and Mercury–Neptune are both blocked by the 0.10 AU sphere. This happens when Neptune is within about 5.7° of Earth’s opposition, about 3.2% of alignments, roughly 200 blinded windows over 2,000 years. For each window we compute the two-leg relay loss through Venus, Mars, Ceres, and Jupiter with exact retarded geometry.

Table 22: E4 blinded-window analysis. Representative Earth–Mercury conjunctions with Neptune near opposition.

t (days)	Venus loss	Mars loss	Ceres loss	Winner
1917.79	46.3%	46.5%	49.3%	Venus (quadrature)
2500.00	58.2%	47.1%	47.0%	Mars
3100.00	46.8%	47.0%	49.5%	Venus
3700.00	47.0%	46.9%	47.1%	Mars (slightly)
4400.00	46.5%	48.0%	49.0%	Venus
5100.00	55.1%	47.5%	47.2%	Ceres

Venus wins 62% of windows, Mars 26%, Ceres 9%, and the relays cover the rare remaining 3%. The two-of-three quorum means the exchange is never fully isolated in the 2000-year scan. Under the path-election rule, Venus is automatically the initiator in every blinded window, because Earth and Mercury have infinite estimated latency to the outer target.

8.4 Analytical bound

Direct loss at the maximum Earth–Neptune distance:

$$d_{\max} = 1.000 + 30.069 = 31.069 \text{ AU}, \quad p_{\max} = 1 - e^{-0.08 \times 31.069} = 0.9167.$$

Single-copy direct success at worst geometry: 0.0833. The bound is stated clearly and used in place of a full scan for the worst case.

9 E5. Shifted Epochs

Tier claimed: **Tier 3**. We test shifts of 1, 10, 50, and 100 Julian years. At each shift, positions advance by the shift, relays advance by the same shift, and the opening balance sheet is unchanged.

Table 23: E5 shifted epochs. Earth–Mars route delay at $h = 0$ and $h = 240$, with 24 h sampled availability.

Offset	$h = 0$ E→M / M→E	$h = 240$ E→M / M→E	Route	Avail.
+1 y	47.185 / 47.183 min	48.288 / 48.286 min	via B	100% / 100%
+10 y	41.948 / 41.947 min	41.984 / 41.982 min	via B	100% / 100%
+100 y	38.503 / 38.501 min	37.434 / 37.431 min	via B	100% / 100%

9.1 Difficult epoch: $h = 650,213$ from base = +74.174 Julian years

At the selected hard phase, $h = 0$ point connectivity exists through Relay B, but the Neptune–B–Earth route is near the 0.10 AU solar boundary and has 0% fixed-path launch availability in the next 24 h sample. The quorum suspends a new remote fill rather than treating the single open instant as a service window. At $h = 240$ Earth–A–Neptune is clear, 285.315 min one way, with 100% sampled availability; Neptune–A–Earth is 285.325 min. The comparable Mars–Earth route at $h = 240$ is Mars–A–Earth, 46.986 min, 100% sampled.

9.2 Hard-phase value move

Amina (Earth) buys 50 Ares shares from Esha (Neptune) at \$100 at $h = 240$. Both lock; fee \$0.50 to the quorum. For the Earth–Neptune pair at $h = 240$, $T(\text{Earth}) = 9.51 \text{ h}$, $T(\text{Venus}) = 9.34 \text{ h}$, $T(\text{Mercury}) = 9.68 \text{ h}$; Venus is elected. The quorum commits after Neptune’s PREPARE at $h = 244.755$; Earth receives the commit at that time; Neptune applies at about $h = 249.510$. End: Amina cash \$144,999.50, shares 650; Esha cash \$75,000, shares 750; quorum \$0.50; others unchanged. Total cash \$500,000; shares 5,000. Conditional completion about 9.510 h from first lock.

9.3 Hard-phase future

Amina long / Esha short 20 contracts, multiplier 100, $P_0 = \$100$, cap $\pm 30\%$, \$60k margin each. Position opens at $h = 244.755$ after Esha’s local lock certificate reaches the quorum; matures at $h = 484.755$. At $P = \$130$ Amina ends \$209,980, Esha \$10,000, quorum \$20; at $P = \$70$ Amina \$89,980, Esha \$130,000, quorum \$20; other accounts \$280,000 in either run. Total \$500,000. Earth–Neptune final payment route at maturity is 285.903 min; remote spendability about $h = 489.521$. Direct advisory successes at opening, $h + 120$, and maturity are 9.128%, 9.189%, 9.249%; if any bulletin is missing, the signed backbone outcome still settles. Claim E5 Tier 3: both funded price directions shown.